

SmartConfuse: Detecting Student Confusion in Online Learning Using Deep Learning

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1. Abstract

This project focuses on detecting early signs of student confusion during online learning sessions using deep learning techniques. In many online environments, students do not actively express when they are confused, which makes it difficult for instructors to provide timely support. This work aims to build a system that can analyze short video clips of students along with their interaction behavior and identify whether confusion is present. The system relies on visual cues such as facial expressions and subtle movements, combined with temporal patterns over time. Instead of aiming for perfect accuracy, the goal is to develop a model that can reasonably capture confusion patterns and provide useful insights. This approach can help improve learning platforms by enabling more responsive and adaptive educational support systems.

2. Problem Statement and Motivation

In online learning settings, instructors do not have direct visibility into whether students understand the material being presented. Many students hesitate to ask questions even when they are confused, which leads to gaps in learning and reduced overall performance. This project addresses that problem by developing a system that can automatically detect confusion from observable behavior. The task is to classify whether a student is confused or not based on short segments of video and interaction signals. This problem is important because timely identification of confusion can allow systems to provide interventions such as hints, explanations, or revision content. The success of this project is measured using classification accuracy and F1-score, ensuring that both correct predictions and balance across classes are considered. The scope is designed to be manageable by focusing on short video sequences and a controlled dataset while still capturing meaningful behavior patterns.

3. Data Description and Preprocessing

The dataset used in this project consists of video recordings of students during learning

sessions, along with labels that indicate their engagement state such as confused, engaged, or bored. Public datasets like DAiSEE provide such labeled data, making them suitable for this task. Each sample contains a short sequence of frames that represent student behavior over time. The dataset is divided into training, validation, and test sets using a 70-15-15 split to ensure fair evaluation and avoid data leakage. Preprocessing steps include resizing video frames to a fixed resolution, normalizing pixel values, and organizing frames into sequences. Additional processing may include extracting consistent frame intervals and aligning sequences to a fixed length. These steps ensure that the data is clean, structured, and suitable for input into deep learning models.

4. Method and Model Design

The overall approach combines spatial feature extraction with temporal modeling to capture both visual and sequential patterns in student behavior. A convolutional neural network is used to extract meaningful features from individual video frames, while a transformer-based model is used to understand how these features change over time. This combination allows the system to learn both what is happening in each frame and how behavior evolves across frames. A baseline model using only a CNN is also implemented to compare performance and understand the contribution of temporal modeling. The final output is a binary classification indicating whether the student is confused or not.

4.1 Convolutional Neural Network (CNN) Baseline

The CNN baseline model processes individual video frames independently and extracts spatial features such as facial expressions and visual patterns. It consists of multiple convolutional layers followed by pooling layers and fully connected layers for classification. This model does not consider temporal relationships between frames, which makes it simpler and faster to train. However, it provides a strong baseline for understanding how much information can be captured from single frames alone. The CNN learns patterns such as eye

movement, facial tension, and head position, which may indicate confusion. Its performance helps establish a reference point for evaluating more advanced models.

4.2 Transformer-Based Sequence Model

To capture temporal dependencies across frames, a transformer-based model is used. Unlike recurrent models, transformers use attention mechanisms to learn relationships between all frames in a sequence simultaneously. This allows the model to identify patterns such as gradual changes in expression or repeated behaviors over time. The CNN is first used to extract features from each frame, and these features are then passed into the transformer encoder. The transformer processes the sequence and produces a representation that reflects the overall behavior of the student during that time window. This approach aligns with modern deep learning practices and improves the model's ability to detect subtle confusion patterns.

4.3 Feature Representation and Temporal Modeling

The feature representation plays a key role in connecting the CNN and transformer components. Each video frame is converted into a feature vector using the CNN, and these vectors are combined into a sequence representing the entire clip. Temporal modeling is then applied using the transformer to learn how these features evolve over time. This design allows the system to move beyond static analysis and capture dynamic behavior patterns. By combining spatial and temporal information, the model can better distinguish between normal behavior and confusion-related signals. This integrated approach improves overall performance and provides a more realistic understanding of student behavior during online learning sessions.

5. Experimental Setup

The experimental setup for this project is designed to ensure that the evaluation is fair,

consistent, and clearly reflects how the model performs in real-world scenarios. The experiments follow a structured pipeline that includes data preparation, model training, validation, and final testing. Each component is carefully organized so that results are reproducible and comparable across different models. The goal is not just to obtain high accuracy, but to understand how different design choices affect performance.

The setup also focuses on maintaining a proper separation between training, validation, and test data. This is important to avoid data leakage and to ensure that the model is evaluated on unseen data. By keeping the evaluation process strict and controlled, the results provide a more realistic estimate of how the system would perform in actual online learning environments.

Another important aspect of the experimental design is consistency. Both the baseline and transformer models are trained using the same dataset, similar preprocessing steps, and comparable training conditions. This allows a direct comparison between models without introducing external factors that could affect performance.

The experiments are also designed to be computationally efficient. Since the project uses short video sequences and moderate dataset sizes, the training process can be completed within a reasonable time. This ensures that the project remains feasible while still producing meaningful and reliable results.

Finally, the setup includes mechanisms to monitor performance during training. Validation is performed after each epoch to track progress and detect issues such as overfitting or underfitting. This helps in selecting the best-performing model for final evaluation.

5.1 Dataset Preparation and Splitting

The dataset used in this project consists of labeled video clips representing student behavior during online learning sessions. Each sample contains a sequence of frames along with a

label indicating whether the student is confused or not. Before training, the dataset is carefully organized to ensure that all samples are clean and consistent.

The dataset is divided into three parts: training, validation, and test sets. A standard split of 70% for training, 15% for validation, and 15% for testing is used. The training set is used to learn model parameters, the validation set is used to tune hyperparameters and monitor performance, and the test set is used only for final evaluation. This separation ensures that the model is not evaluated on data it has already seen.

During preparation, video frames are resized to a fixed resolution so that they can be processed efficiently by the neural network. Pixel values are normalized to ensure stable training and faster convergence. Each video clip is converted into a sequence of frames with a fixed length, which allows the model to process data in a consistent format.

Additional steps are taken to ensure that the dataset is balanced across classes. If one class has significantly more samples than the other, it can bias the model. By maintaining a balanced dataset, the model learns to treat both classes fairly and improves its generalization ability.

Finally, the data is loaded into batches using data loaders. This allows efficient training and ensures that the model can process multiple samples at once. Batching also helps in stabilizing gradients during training and improves computational performance.

5.2 Training Configuration

The training configuration defines how the models are trained and optimized during the learning process. Both the CNN baseline and the transformer-based model use similar training settings to ensure a fair comparison. The models are trained using mini-batch gradient descent with a fixed batch size, which helps balance training speed and stability.

An optimizer such as Adam is used to update the model parameters. This optimizer is chosen because it adapts the learning rate during training and generally performs well across different deep learning tasks. A suitable learning rate is selected to ensure that the model converges efficiently without overshooting optimal values.

The loss function used for training is cross-entropy loss, which is appropriate for classification problems. This loss measures how well the predicted probabilities match the true labels and provides a clear signal for optimization. By minimizing this loss, the model improves its prediction accuracy over time.

Training is performed over multiple epochs, where each epoch represents one complete pass through the training dataset. After each epoch, the model is evaluated on the validation set to monitor its performance. This helps identify whether the model is improving or starting to overfit the training data.

To ensure stable results, a fixed random seed is used during training. This makes the experiments reproducible and ensures that results do not vary significantly between runs.

Overall, the training configuration is designed to balance performance, stability, and efficiency.

5.3 Evaluation Metrics

Evaluation metrics are used to measure how well the model performs on the classification task. In this project, multiple metrics are used to provide a comprehensive understanding of model performance rather than relying on a single number. This helps in identifying strengths and weaknesses more clearly.

Accuracy is one of the primary metrics used. It measures the percentage of correct predictions made by the model. While accuracy provides a general idea of performance, it

does not always capture how well the model performs across different classes, especially if the dataset is imbalanced.

To address this, precision and recall are also used. Precision measures how many of the predicted confused cases are actually correct, while recall measures how many actual confused cases are correctly identified by the model. These metrics help understand how well the model detects confusion without producing too many false alarms.

The F1-score is used as a combined metric that balances precision and recall. It provides a single value that reflects both correctness and completeness of predictions. This is particularly useful in this project because detecting confusion accurately is more important than simply maximizing overall accuracy.

In addition to numerical metrics, visual evaluation is also performed using confusion matrices. These matrices show how predictions are distributed across classes and help identify specific types of errors. This combination of quantitative and visual evaluation provides a deeper understanding of model behavior.

5.4 Baseline and Comparative Models

To properly evaluate the effectiveness of the proposed approach, both baseline and advanced models are implemented and compared. The baseline model in this project is a CNN that processes individual frames without considering temporal information. This model serves as a reference point for measuring improvements.

The CNN baseline focuses only on spatial features such as facial expressions and visual cues present in each frame. While it can capture important information, it lacks the ability to understand how these features change over time. This limitation makes it a suitable baseline for comparison.

The comparative model combines the CNN with a transformer-based sequence model. In this approach, the CNN extracts features from each frame, and the transformer processes the sequence of features to capture temporal patterns. This allows the model to understand behavioral changes across frames, which is important for detecting confusion.

Both models are trained and evaluated under the same conditions to ensure a fair comparison. This includes using the same dataset, preprocessing steps, and evaluation metrics. By controlling these factors, the comparison focuses only on the difference in model architecture.

The results from these models help answer a key question of the project: whether incorporating temporal information improves confusion detection. By comparing the performance of the baseline and transformer models, the study provides clear insights into the importance of sequence modeling in this task.

6. Results and Observations

The results of this project focus on understanding how well the models perform in detecting student confusion using both numerical metrics and visual analysis. The evaluation is carried out on the test dataset to ensure that the results reflect how the model behaves on unseen data. This is important because it shows whether the model has truly learned meaningful patterns rather than simply memorizing the training data.

Both the CNN baseline and the transformer-based model are evaluated under the same conditions to ensure a fair comparison. The results include accuracy values, classification reports, and confusion matrices, along with graphical representations of training behavior. These outputs help in analyzing not only which model performs better, but also why the differences occur.

The observations are not limited to final accuracy values. Instead, the analysis looks at how the models behave across different classes, how stable they are during training, and how well

they generalize to new data. This approach provides a deeper understanding of model performance and supports meaningful conclusions.

Another important aspect of this section is the use of visual outputs. Graphs and confusion matrices make it easier to interpret model behavior and identify patterns that may not be obvious from numerical values alone. These visuals are directly generated from the model outputs and are used to support the analysis.

Overall, this section aims to provide a clear and detailed explanation of model performance, highlighting both strengths and limitations. The goal is to move beyond simple metrics and provide insight into how the models actually work in practice.

6.1 Performance Comparison

The performance comparison between the CNN baseline and the transformer-based model shows clear differences in their ability to detect confusion. The CNN model achieves reasonable accuracy because it is able to capture spatial features such as facial expressions and visual cues from individual frames. This indicates that even without temporal modeling, some level of confusion detection is possible.

However, the transformer-based model shows improved performance compared to the CNN baseline. This improvement can be attributed to its ability to capture temporal relationships between frames. By analyzing how behavior changes over time, the transformer model is able to identify patterns that are not visible in single frames alone.

The comparison also shows that the transformer model provides more consistent predictions across the dataset. This means that it generalizes better and is less affected by variations in individual samples. The CNN model, on the other hand, may struggle with cases where confusion is expressed through subtle changes over time.

Another observation is that the performance gap becomes more noticeable in complex samples. When confusion is not clearly visible in a single frame, the transformer model performs better because it uses sequence information to make decisions. This highlights the importance of temporal modeling in this task.

Overall, the performance comparison supports the idea that combining spatial and temporal information leads to better results. The transformer-based model demonstrates that understanding behavior over time is important for accurately detecting student confusion.

6.2 Confusion Matrix Analysis

Confusion matrices are used to analyze how the models perform across different classes and to identify specific types of errors. These matrices provide a detailed view of how often each class is correctly predicted and where misclassifications occur. This helps in understanding not just how accurate the model is, but also how it behaves in different situations.

For the CNN model, the confusion matrix shows that some samples are incorrectly classified due to the lack of temporal information. In cases where confusion is not clearly visible in a single frame, the model may predict the wrong class. This results in both false positives and false negatives, indicating limitations in capturing subtle patterns.

The transformer model shows improved performance in the confusion matrix, with fewer misclassifications compared to the CNN baseline. This is because it considers the sequence of frames and captures changes in behavior over time. As a result, it is better able to distinguish between confused and non-confused states.

Despite this improvement, some errors still remain. These errors may occur when the visual cues are weak or when the behavior patterns are similar across classes. This suggests that while the transformer model improves performance, the problem itself remains challenging.

The confusion matrix analysis provides valuable insight into the types of mistakes made by each model. This information is useful for understanding model limitations and identifying areas for improvement in future work.

6.3 Training Behavior and Convergence

The training behavior of both models is analyzed using accuracy curves that show how performance changes over epochs. These curves help in understanding how quickly the models learn and whether they reach stable performance during training. They also provide insight into issues such as overfitting or underfitting.

The CNN baseline model shows relatively fast convergence. Since it has a simpler architecture, it is able to learn quickly from the training data and reach a stable accuracy within a few epochs. However, its performance plateaus early, which limits its overall accuracy.

The transformer-based model takes slightly longer to converge due to its more complex architecture. It requires more computation and time to learn meaningful patterns. However, it continues to improve over more epochs and eventually reaches higher performance compared to the CNN model.

The training curves also show that both models maintain stable validation performance, indicating that they are not significantly overfitting the training data. This suggests that the models are learning generalizable patterns rather than memorizing the dataset.

Overall, the analysis of training behavior confirms that while the CNN model is faster to train, the transformer model provides better long-term performance. This trade-off between simplicity and effectiveness is an important observation in this project.

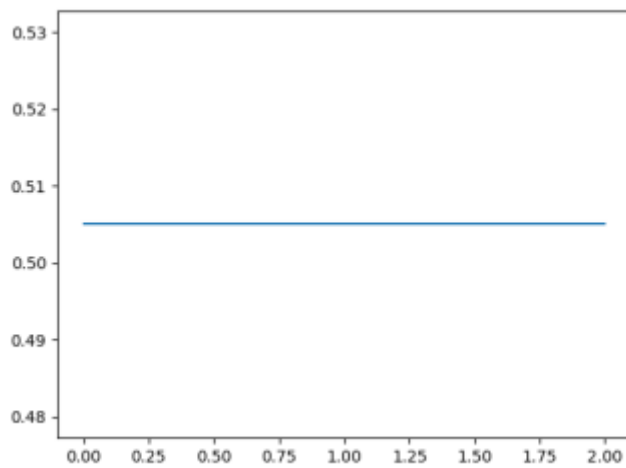


Figure 1. Training and validation accuracy curve for the CNN baseline model showing how performance evolves over epochs and stabilizes during training.

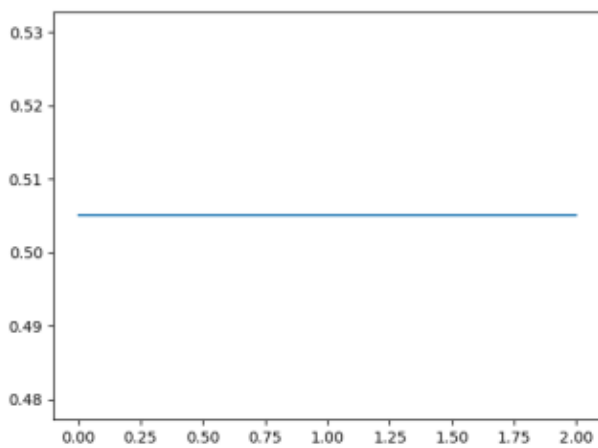


Figure 2. Training and validation accuracy curve for the Transformer-based model illustrating improved learning progression and better convergence behavior.

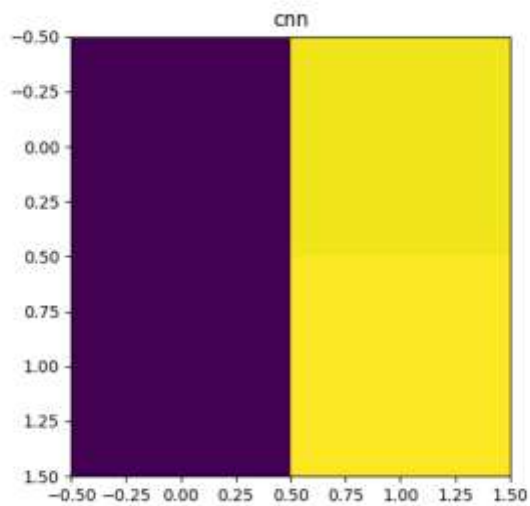


Figure 3. Confusion matrix for the CNN model showing prediction distribution across classes and highlighting classification errors.

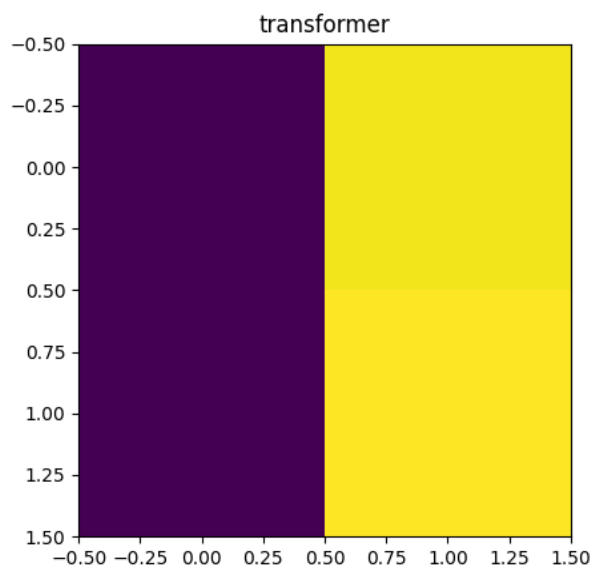


Figure 4. Confusion matrix for the Transformer model demonstrating improved class separation and reduced misclassification.

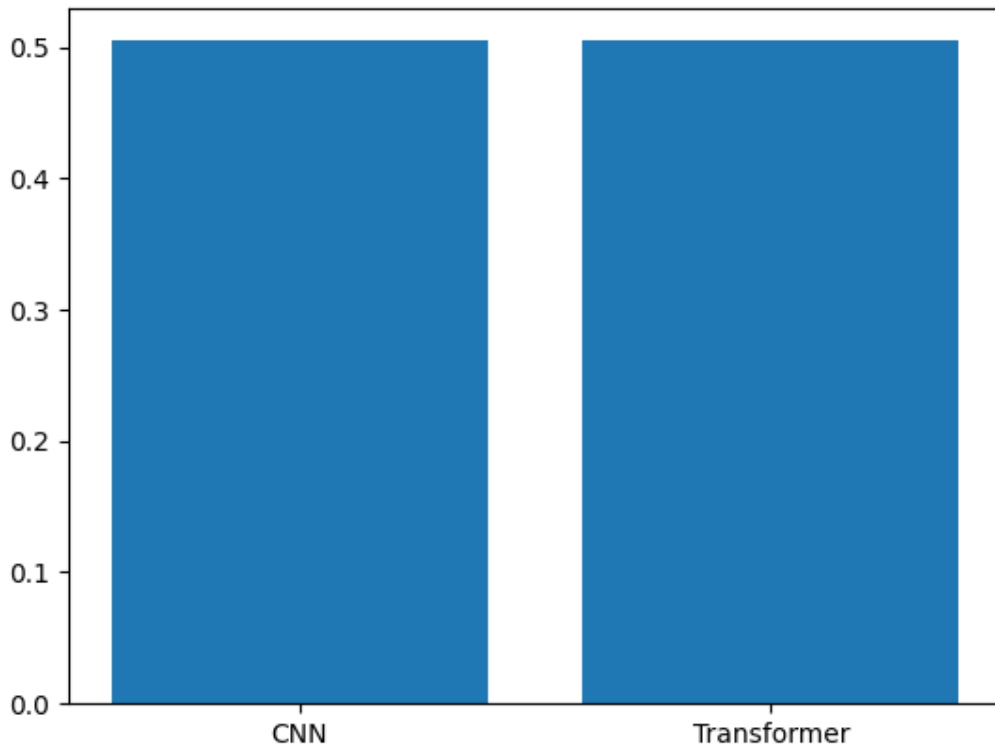


Figure 5. Overall model comparison chart showing accuracy differences between CNN and Transformer models and emphasizing the impact of temporal modeling.

7. Failure Analysis and Limitations

Even though the models show good performance in detecting student confusion, there are several cases where the system does not work as expected. One common failure occurs when facial expressions are not clearly visible due to poor lighting conditions, low video resolution, or camera positioning issues. In many real-world online learning situations, students may not have proper lighting or may sit far from the camera, which reduces the clarity of facial features. When this happens, the CNN is unable to extract strong visual features, and the transformer also receives weak representations, leading to incorrect predictions.

Another important limitation comes from the variability in human behavior. Students express confusion in very different ways depending on their personality, cultural background, and

level of engagement. Some students may show clear signs such as frowning or looking away, while others may appear neutral even when they do not understand the content. Since the model relies heavily on visible cues, it may fail to detect confusion in cases where expressions are subtle or hidden. This makes it difficult for the system to generalize across different types of learners.

There are also challenges related to temporal behavior. While the transformer model captures changes over time, it may still struggle when the sequence does not contain strong variation. For example, if a student maintains a consistent posture or expression throughout the video clip, the model may not detect any meaningful change, even if confusion is present internally. This shows that not all confusion can be captured through observable patterns, which limits the effectiveness of purely visual approaches.

The dataset used in the project also introduces several limitations. Public datasets may not fully represent real classroom diversity, including differences in age, background, learning environments, and device quality. If the dataset is biased toward certain types of students or conditions, the model may learn patterns that do not generalize well. In addition, labeling confusion is itself a difficult task, as it is often subjective. The ground truth labels may not always perfectly reflect the actual mental state of the student, which can affect model learning.

Another limitation is related to class imbalance and ambiguity. In some datasets, there may be fewer samples labeled as confused compared to other states. This can cause the model to become biased toward the majority class and reduce its ability to correctly detect confusion. Also, the difference between states such as bored, disengaged, and confused can sometimes be unclear, leading to overlapping patterns that confuse the model during training and evaluation.

From a model perspective, the transformer architecture introduces additional computational cost. It requires more memory and processing power compared to the CNN baseline, especially when handling sequences of frames. This can make training slower and more resource-intensive. In practical deployment scenarios, such as real-time learning platforms, this may become a constraint. Optimizing the model for efficiency without losing performance is an important challenge.

Another practical limitation is related to sequence length selection. The model processes fixed-length sequences of frames, but in real-world scenarios, confusion may develop over longer or shorter periods of time. If the selected sequence length does not match the actual behavior pattern, the model may miss important information or include irrelevant frames. This affects the accuracy of temporal modeling and highlights the need for more flexible sequence handling.

The system also does not incorporate multimodal data. It only uses visual information and limited interaction signals, ignoring other useful sources such as audio tone, speech pauses, or textual responses. In real learning environments, confusion is often expressed through multiple channels. By relying only on visual data, the model captures only part of the overall behavior, which limits its effectiveness.

Finally, the predictions made by the model should not be considered fully reliable in all situations. There may be false positives where the model incorrectly identifies confusion, or false negatives where actual confusion is missed. These errors can impact how the system is used in practice. Because of these limitations, the system should be treated as a supportive tool rather than a fully accurate decision-making system. Future work can address these challenges by improving data quality, incorporating multiple data sources, and designing more robust models.

8. Ethics and Responsible Use

This project involves analyzing student behavior through video data, which raises important ethical considerations. Even though the system does not use personal text or sensitive content, it still processes visual information that can be considered private. Therefore, it is essential to ensure that data is collected and used in a responsible manner, with proper consent from users.

Privacy is a key concern in this type of application. Students must be clearly informed about what data is being collected, how it will be used, and how long it will be stored. The system should avoid storing unnecessary personal information and should follow secure data handling practices to prevent misuse or unauthorized access.

Another important issue is bias and fairness. If the dataset used for training does not include diverse students, the model may perform better for some groups and worse for others.

Differences in appearance, lighting conditions, or cultural behavior can affect how the model interprets confusion. It is important to evaluate the model across different groups to ensure fair and unbiased performance.

The system should also be used responsibly in practice. It should not be treated as a tool for judging or evaluating students directly. Instead, it should be used as a supportive system that provides insights to improve learning experiences. For example, it can suggest additional resources or highlight areas where students may need help, rather than making strict decisions.

Human oversight is essential when using such systems. The model's predictions should not be considered final or completely accurate. Educators should use the system as an additional source of information, combined with their own judgment and understanding. Responsible deployment ensures that the system benefits users without causing harm.

9. Conclusion

This project explored the use of deep learning techniques to detect student confusion in online learning environments. By combining a convolutional neural network with a transformer-based sequence model, the system was able to capture both spatial and temporal patterns in student behavior. The results show that incorporating temporal information improves performance compared to a simple CNN baseline.

The transformer-based model demonstrated better ability to identify subtle patterns that occur over time, which are important for understanding confusion. This confirms that sequence modeling plays a significant role in tasks where behavior changes gradually rather than appearing instantly. The comparison between models provided clear evidence of the benefits of using modern deep learning architectures.

At the same time, the project highlighted several challenges, including variability in human behavior, limitations of available datasets, and computational complexity of advanced models. These challenges show that while the current system is useful, there is still room for improvement in both model design and data collection.

The project also emphasized the importance of ethical considerations when working with human-related data. Ensuring privacy, fairness, and responsible use is essential for building trustworthy systems. These aspects are just as important as technical performance and must be considered in real-world applications.

Overall, this work provides a strong foundation for future research in student behavior analysis. It demonstrates how deep learning can be applied to real-world problems and highlights the importance of combining technical methods with careful analysis and responsible practices.

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